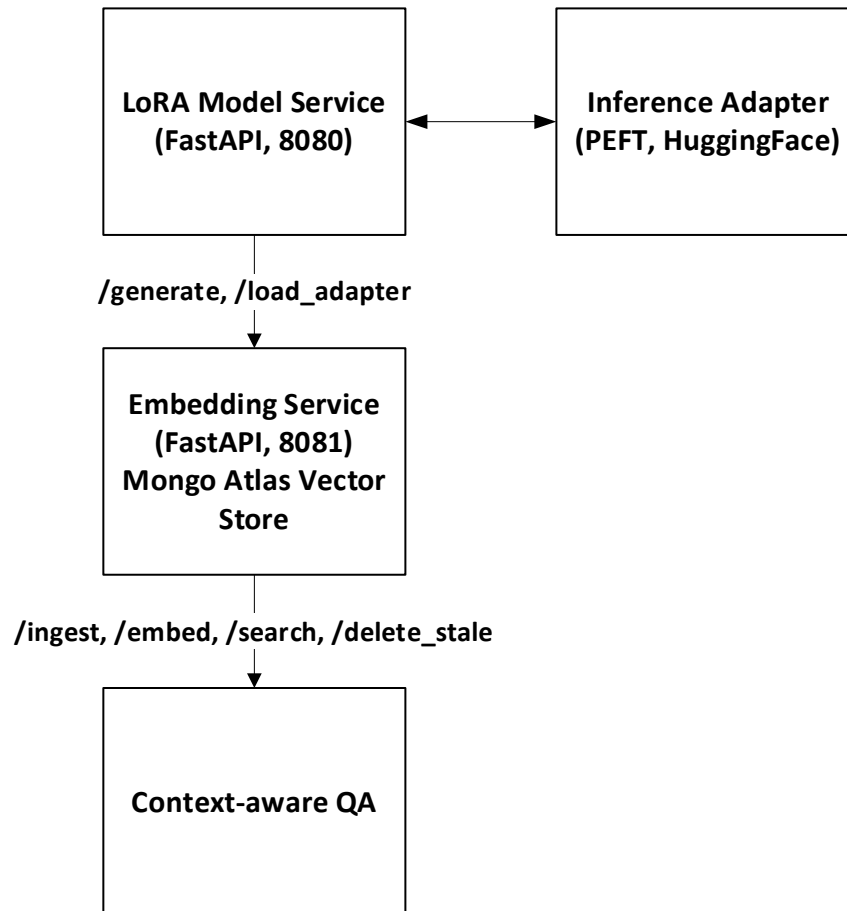


Cover Note – LoRA + Embedding Services

Overview

This repo implements a **prototype context-aware QA system** with two main services:



- **LoRA Model Service:** applies PEFT adapters to base models for text generation.
- **Embedding Service:** ingests documents, creates embeddings, and performs vector search.

Assumptions

- Base model: distilgpt2, LoRA adapters applied locally.
- Mongo Atlas used for vector storage.
- GPU optional but recommended.
- Default similarity metric: cosine.

Shortcuts / Simplifications

- No automated CI/CD, manual testing via PowerShell / curl.
- Minimal logging, no Prometheus metrics.
- No React frontend, demo scripts only.
- Multi-vector metric support is not fully configurable.
- Deployment guidance is high-level, the AWS runbook is incomplete.

Incomplete / Future Work

- CI/CD workflow with lint + unit tests + Docker builds.
- Metrics for request latency & error counts.
- React frontend for `/chat` + source display.
- Runtime selection of similarity metric (cosine/dot).

- Detailed AWS deployment/runbook.
 - Support for larger models and sampling options for generation.
-

Notes

The prototype demonstrates **core integration** of LoRA adapters with embeddings and vector search. Production concerns (auth, scaling, caching, observability) are not fully addressed.